

Real Time Face Attendance System

ABSTRACT: One's face is a physical depiction of who they are. As a result, we have suggested a facial recognition-based automatic student attendance system. The usage of face recognition technology in everyday life is particularly beneficial for security control systems. Face recognition is used by the FBI (Federal Bureau of Investigation) for criminal investigations as well as the airport security system to identify criminals. In the first step of our suggested method, video framing, the camera is activated using a user-friendly interface. The Viola-Jones method is used to identify and segment the face ROI from the video frame. If necessary, image size scaling is done at the pre-processing step to prevent information loss. After using the median filter to reduce noise, color pictures are transformed into grayscale versions. Then, to improve the contrast of the pictures, contrast-limited adaptive histogram equalization (CLAHE) is applied. Principal component analysis (PCA) and improved local binary pattern (LBP) are used appropriately in the face identification stage to extract the characteristics from facial pictures. By minimizing the lighting impact and boosting identification rates, the improved local binary pattern outperforms the original local binary pattern in our suggested method. The features that were retrieved from the training pictures and the features that were extracted from the test images are then compared. The face pictures are then identified and categorized using the upgraded LBP, PCA, and algorithm that produced the best results.

The acknowledged student's attendance will then be recorded and saved in the excel file. A warning will be sent if a student signs in more than once. The student who is not enrolled will also be allowed to register right away. When two photographs per person are trained, the average recognition accuracy is 100% for high-quality images, 94.12% for low-quality images, and 95.76% for the Yale face database.

I.INTRODUCTION

The primary goal of this project is to create an automated student attendance system based on

facial recognition. The test pictures and training images of this suggested technique are restricted to frontal and upright facial images that only contain a single face to improve performance. To ensure no quality variation, the test photographs and training images must be taken using the same equipment. To be identified, the pupils must also register in the database. The user-friendly interface allows for immediate enrollment.

LITERATURE SURVEY

The RFID (Radio Frequency Identification) card system, the fingerprint system, and the iris recognition system all have drawbacks, according to Arun Katara et al. (2017). The RFID card method is used because it is straightforward. If the user has their friend's ID card, they usually assist their pals with checking in. The fingerprint technology is efficient but not very quick because each user must stand in line and go through the verification procedure individually. The human face, however, is constantly visible and contains less data than the iris when used for facial identification. A more detailed iris recognition technology could violate the user's privacy. Although voice recognition is a possibility, it is not as precise as other approaches. Thus, it is advised that the student attendance system be upgraded to include face recognition technology.

SYSTEM ANALYSIS

A. Existing Systems:

While numerous real-time face attendance systems prioritize high accuracy, our project using the Viola-Jones algorithm focuses on maximizing speed and efficiency for student attendance scanning within seconds. This approach leverages pre-trained models, eliminating extensive data collection and training, potentially addressing privacy concerns in educational settings. This focus on speed, efficiency, and privacy complements existing high-accuracy systems, making our Viola-Jones based solution ideal for situations where swift student attendance verification and data privacy are paramount.

METHODOLOGY

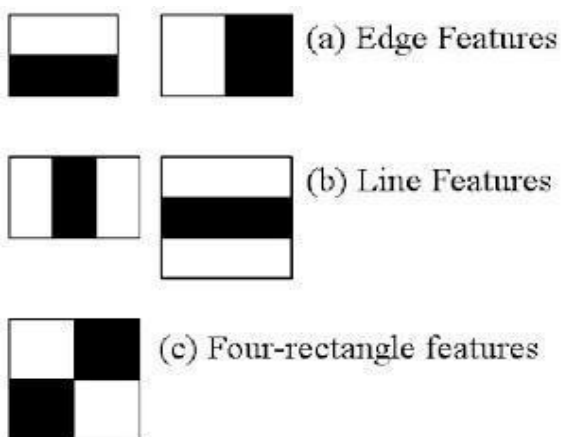
The method uses a facial recognition-based attendance system for students. The technique starts with the picture capturing utilizing an easy-to-use interface, then moves on to pre-processing the facial photos that were collected, feature extraction from those images, subjective selection, and finally categorization of the facial images to be recognized. This suggested technique computes both LBP and PCA feature extraction algorithms, then compares their performance. With this method, LBP is improved to lessen the lighting impact. Moreover, a subjective selection method that combines improved LBP and PCA is created to improve accuracy. The next sections will go through the specifics of each level.

Input Images:

Real-time face recognition student attendance systems should utilize our own database, but the databases given by earlier researchers are also used to develop the system more effectively, efficiently, and for assessment.

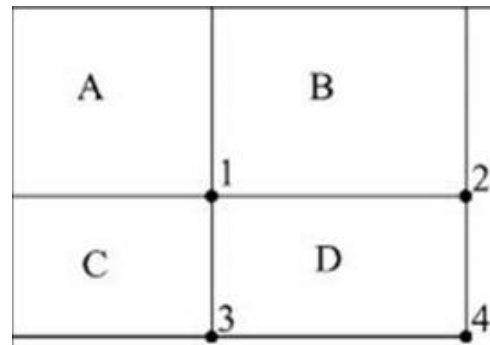
To assess the performance, the Yale face database is employed as both a training set and a testing set. One hundred and sixty-five grayscale pictures of fifteen people may be found in the Yale face database. There are eleven photos for everyone, and each one is in a distinct state. The situations included right-light, sad, drowsy, astonished, and wink. They also included center-light, with glasses, joyful, left-light, without glasses, normal. The database's several variants enable the system to function consistently under a wide range of circumstances and settings.

Algorithm: The most widely used technique to locate the face segment from a static picture or video frame is the Viola-Jones algorithm, which was developed by P. Viola and M. J. Jones in 2001. The Viola-Jones algorithm is essentially composed of four pieces. The first component is called the Haar feature, the second component is where the integral picture is formed, the third component is where Adaboost is implemented, and the fourth component is the



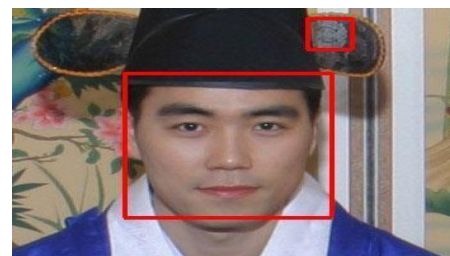
cascade process

given image is examined using the Viola-Jones method utilizing Haar features, which are made up of several rectangles (Mekha Joseph et al., 2016). Several Haar feature types are displayed in Figure 2.1. The attributes serve as window functions that map onto the picture. Subtracting the total of the white rectangle(s) from the total of the black rectangle(s) yields a single number representing each feature (Mekha Joseph et al., 2016).



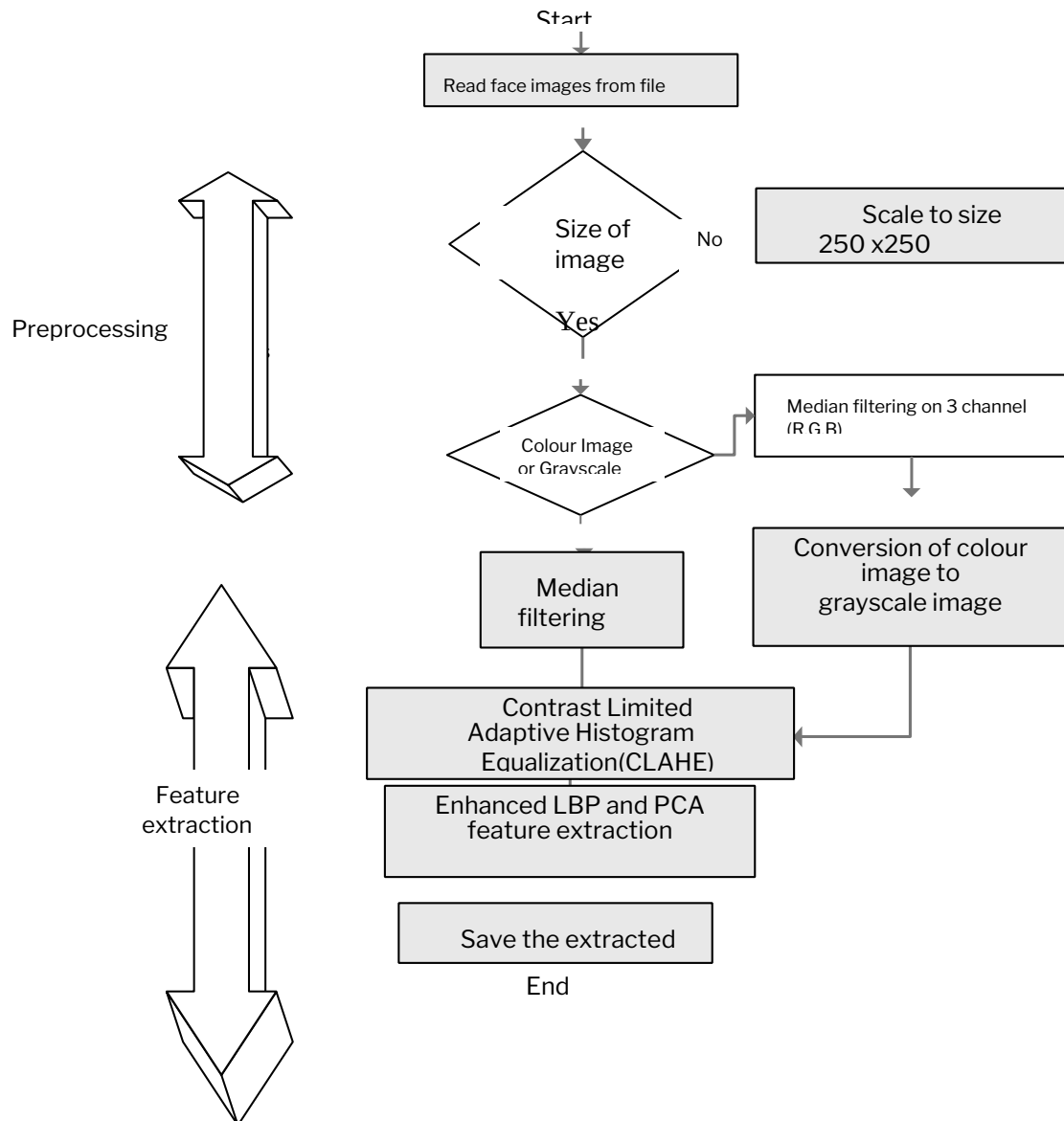
The total of the pixels on the left and the top of the relevant place determines the value of an integrated image in that position. The value of the integral picture at position 1 is the total of the pixels in rectangle A, to provide a clear illustration. The integral image values for the remaining sites add up. For instance, the value at location 2 represents the addition of A and B, $(A + B)$, the addition of A and C, $(A + C)$, and the addition of all the regions, $(A + B + C + D)$, at position 3. (Srushti Girhe et al., 2015). As a result, to exclude rectangles A, B, and C, the total in the D region may be calculated simply using addition and subtraction of the diagonal at point $4 + 1$ $(2 + 3)$.

Adaboost, also known as "Adaptive Boosting," is a well-known boosting strategy in which numerous "weak classifiers" are blended into a single "strong classifier," according to Burak Ozen (2017) and Chris McCormick (2013). According to the output of the previous classifier, the training set is chosen for each new classifier and establishes how much weight should be assigned to each classifier in order to make it relevant. Based on human eyesight, it had to be manually removed since erroneous detection can happen. A fake face detection example is show in figure.



3.1 Data Flow Diagram

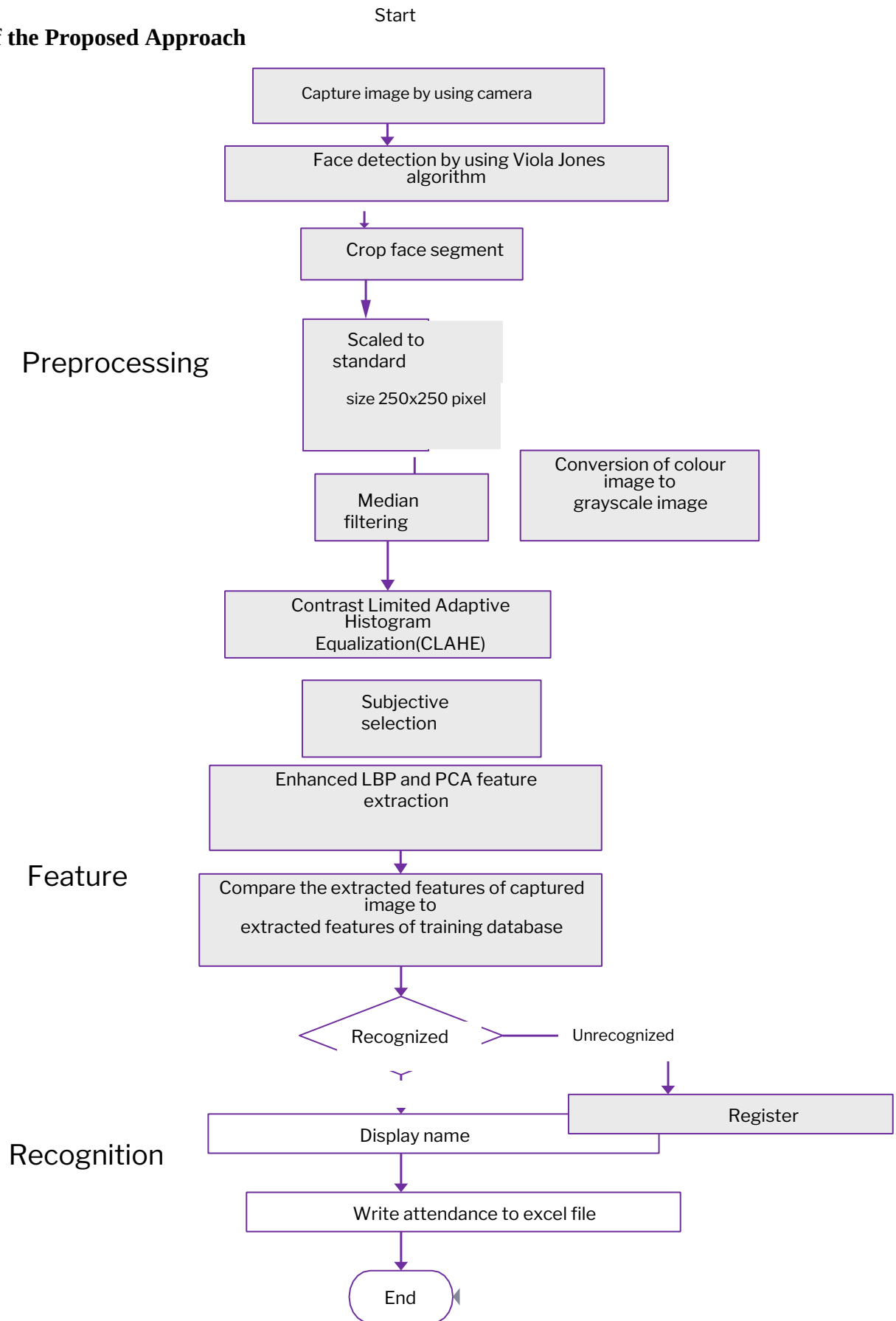
Training database



The input image for the proposed approach must be frontal, upright and only a single face. Although the system is designed to be able to recognize the student with glasses and without glasses, students should provide both facial images with and without glasses to be trained to increase the accuracy to be recognized without glasses. The training image and testing image should be captured by using the same device to avoid quality difference. The students must register in order to be recognized. Enrolment can be done on the spot through the user-friendly interface. These conditions must be satisfied to ensure that the proposed approach can perform well.

The face will be identified from the video camera recording frame using the Viola-Jones object detection framework. Chapter 2 describes the Viola-Jones algorithm's operating system. The Viola-Jones framework is restricted by the requirement that the facial picture be a frontal upright image and that the subject's face in a video frame face the camera.

Flow of the Proposed Approach



Median Filtering:

A reliable technique for noise reduction is median filtering. It is frequently used in many applications because it can reduce unwanted noise while keeping important picture detail. As color photographs taken with a camera are RGB pictures, median filtering is applied to each of the image's three channels. The image before and after noise reduction using median filtering in three channels is shown in Figure . If the source picture is grayscale, the channels don't need to be separated before doing the median filtering.



Conversion to Grayscale Image:

The suggested contrast enhancement method CLAHE can only be used on grayscale photos, even though cameras generate color images. The lighting impact of the photos can be lessened after enhancing contrast. LBP derives the grayscale characteristics as an 8-bit texture descriptor from the contrast-improved pictures (Ojala, T. et al., 2002). As a result, before moving on to the next steps, color photographs must be converted to grayscale images. The complexity of the computation can be lowered by transforming color photos to grayscale images, increasing computing speed (Kanan and Cottrell, 2012). Images are transformed into grayscale pictures in Figure.



Conversion of Image to Grayscale Image

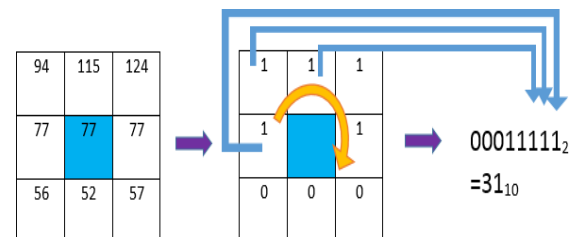
Working Principle of Original LBP:

LBP is essentially a descriptor that uses textures and encodes local primitives as binary strings. Timo Ojala and colleagues, 2002. The 3 3 mask size is what the original LBP

operator uses. 9 pixels are included in a 3 by 3 mask. The nearby pixels (the other 8 pixels) will be converted into binary digits using the central pixel as a threshold. The adjacent pixel value receives a 1 if it is greater than the center pixel value, else it receives a 0. Following that, a binary code made up of the neighboring pixel bits is concatenated to create a byte value that represents the center pixel. LBP conversion example is shown in Figure

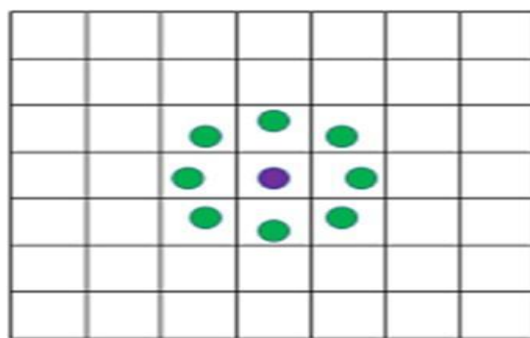
Working Principle of Proposed LBP:

The original LBP operator is made up of a 9-pixel, 3 3 filter size. It appears to be more rectangular in form than round. Every feature, including those that are not crucial,

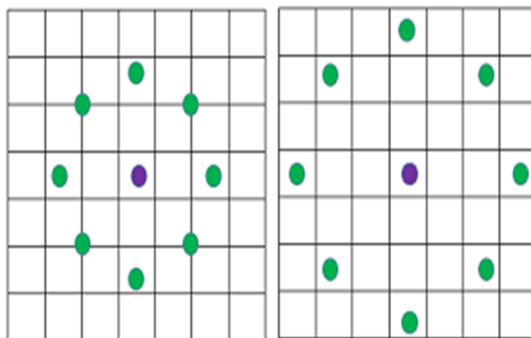


will be sampled since there are 9 pixels close to one another. Due to the tiny filter size's emphasis on small scale detail (Lee and Li, 2007), including the shadow produced by non-uniform lighting condition, it is more impacted by uneven lighting conditions. In our suggested method, the LBP operator is implemented with a bigger radius size, R. The equation for changing the radius size has been introduced in the 2013 work by Md. Abdur Rahim et al. The impact of altering the radius size was not mentioned in the paper, though. To improve the system and lessen the lighting impact, the suggested technique analyses various radius sizes. The filter size will rise when the radius size does as well. R denotes the distance from the center pixel, the angle at which the sampling point was taken, and P the number of sampling points that were taken on the circle's periphery and compared to the center pixel. When the neighboring notation (P, R,) is used, the sines and cosines provided in the equation may be used to compute the coordinates of the center pixel (X_c, Y_c) and the P neighbors (X_p, Y_p) on the edge of the circle with radius R (Md. Abdur Rahim et al., 2013).

Despite the larger radius, a total of 8 sample points are still obtained, which is the same as the original LBP operator. The technique involves performing CLAHE on the input facial photographs in grayscale to increase contrast. The photos with increased contrast are still grayscale. The suggested LBP operator uses just 8 bits of computation to extract grayscale features from contrast-improved grayscale pictures. After that, in a manner similar to the initial LBP operator encoding method, the pixels at the sampling sites will be encoded as an 8-bit binary string. In comparison to the original LBP, enhanced LBP with radius size two performs better and has a more reliable recognition rate. Hence, enhanced LBP with radius size two will

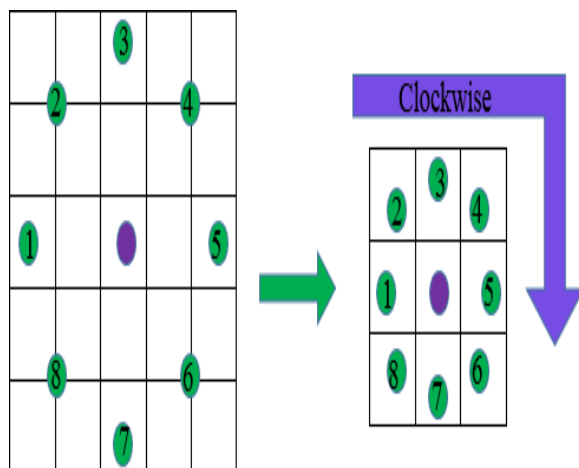


R=1(original)



R=2

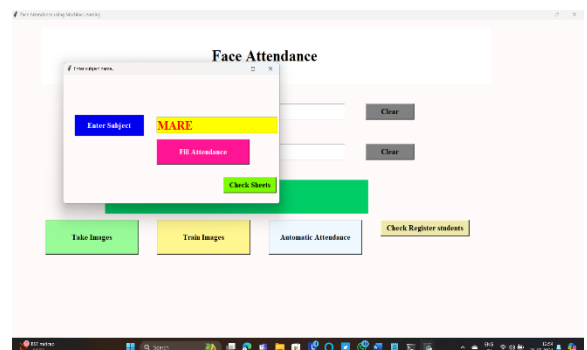
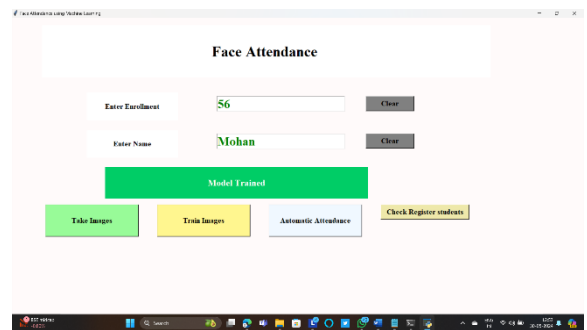
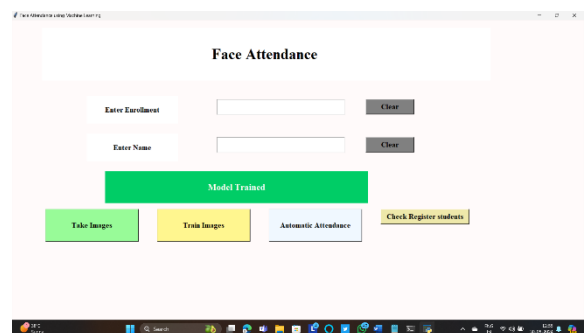
R=3

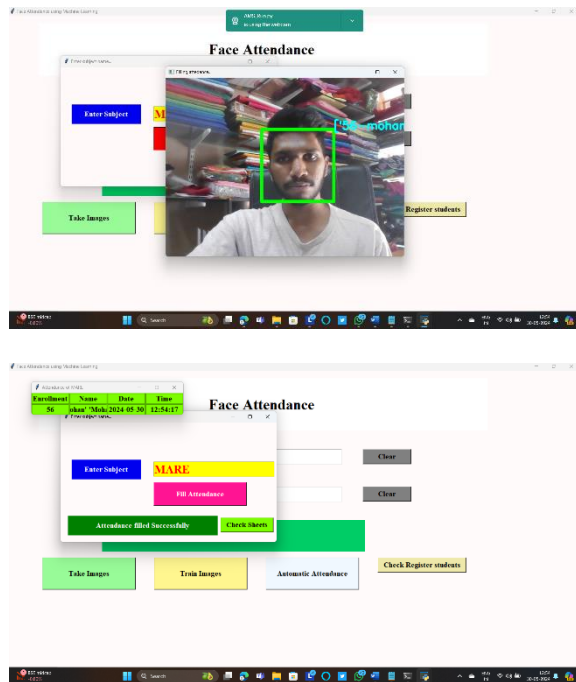


Proposed LBP Operator with Radius 2 and Its Encoding Pattern

Results:

The recommended technique uses the MATLAB GUI to produce a user-friendly face recognition student attendance system (Graphic User Interface). A few buttons are included on the interface, each of which has a specific function. The register button enables student enrollment or registration, the start button initializes the camera and performs facial recognition automatically based on faces identified, and the update button trains the most current photos that have been registered in the database. The browse and recognise buttons are used to browse through face images from a selected database, identify the chosen image, and evaluate the effectiveness of the system. The enhanced LBP with radius two method is used in this part, along with a proposal.





Conclusion:

This method provides a detailed description of an automated student attendance system based on facial recognition. With the help of a comparison between the input picture from a video frame that was recorded and the training image, the suggested strategy offers a way to recognize the persons. This suggested method is capable of locating a face from an input facial picture that is taken from a frame of the recorded video. Moreover, it offers a pre-processing technique to improve image contrast and lessen the lighting impact. Both LBP and PCA are used to extract characteristics from the face picture. The system may be stabilized by producing consistent results thanks to the method that combines LBP and PCA. When two photographs per person are taken, the accuracy of the suggested method is 100% for high-quality images, 92.31% for low-quality images, and 95.76% for the Yale face database.

The extraction of face features may be difficult, particularly in variable lighting, as a conclusion to the investigation. Contrast Limited Adaptive Histogram Equalization (CLAHE), used in pre-processing, can lessen the lighting impact. When it comes to improving contrast, CLAHE outperforms histogram equalization. When compared to the original LBP operator, enhanced LBP with bigger radius sizes—specifically, radius size two—perform better, are less impacted by light, and are more consistent when compared to other radius sizes.

Recommendation:

There are certain restrictions in the suggested technique. The input image must first be frontal and a single, upright face image. Second, if there are severe lighting issues, accuracy may suffer. Thirdly, if the image that was obtained is blurry, incorrect recognition could happen. Moreover, LBP is a descriptor that extracts local characteristics based on textural data. To achieve high accuracy, test and train images must be of the same quality and be taken with the same instrument. Finally, if a person is wearing makeup in a face recognition photo, the key characteristics will be hidden.

A better camera with a stronger light source can lessen the illumination issue and prevent the acquisition of blurry photographs. With the suggested method, the laptop's built-in camera serves as the default device. The laptop camera's illumination, however, is quite weak, which makes the system unsteady. To have better results in the future, a better camera and lighting source can be employed. This can lessen the reliance on ambient brightness, particularly in locations where test and training photos are taken. Moreover, it is possible to construct a face recognition system that recognizes several faces in addition to a single facial picture. This may improve the system's effectiveness. A technique, such as the affine transform, can be used in the pre-processing step to align the face picture based on coordinates in the center of the eyes. This might be beneficial, particularly for the PCA technique, which maps the test picture to the training image to do face recognition. This approach largely depends on the camera that was used to take the photographs; therefore the test image and train image are related to one another. This tactic must make use of the same capturing tools in order to be successful. As a result, alternative algorithms, such as the A.I. (artificial intelligence) algorithm, which may be used for face identification, may take the role of LBP. Convolution Neural Networks (CNNs), a machine deep learning technology that has recently gained popularity, may do recognition with less reliance on a particular train image given a large library. CNN, on the other hand, requires either a very small class size or a very large database to increase its accuracy.

References:

Robinson-Riegler, G., & Robinson-Riegler, B. (2008). *Cognitive psychology: applying the science of the mind*. Boston, Pearson/Allyn and Bacon.

Margaret Rouse. (2012). What is facial recognition? - Definition from WhatIs.com. [online] Available at: <http://whatis.techtarget.com/definition/facial-recognition> [Accessed 25 Mar. 2018].

Solon, O. (2017). Facial recognition database used by FBI is out of control, House committee hears. [online] the Guardian. Available at: <https://www.theguardian.com/technology/2017/mar/27/us-facial-recognition-database-fbi-drivers-licenses-passports> [Accessed 25 Mar. 2018].

Robert Silk. (2017). Biometrics: Facial recognition tech coming to an airport near you: Travel Weekly. [online] Available at: <http://www.travelweekly.com/Travel-News/Airline-News/Biometrics-Facial-recognition-tech-coming-airport-near-you> [Accessed 25 Mar. 2018].

Sidney Fussell. (2018). NEWS Facebook's New Face Recognition Features: What We Do (and Don't) Know. [online] Available at: <https://gizmodo.com/facebooks-new-face-recognition-features-what-we-do-an-1823359911> [Accessed 25 Mar. 2018].

deAgonia, M. (2017). Apple's Face ID [The iPhone X's facial recognition tech explained]. [online] Computerworld. Available at: <https://www.computerworld.com/article/3235140/apple-ios/apples-face-id-the-iphone-xs-facial-recognition-tech-explained.html> [Accessed 25 Mar. 2018].

Ashley DuVal. (2012). Face Recognition Software -History of Forensic Psychology. [online] Available at: <http://forensicspsych.umwblogs.org/research/criminal-justice/face-recognition-software/> [Accessed 12 Apr. 2018].

Jesse Davis West. (2017). History of Face Recognition - Facial recognition software. [online] Available at: <https://www.facefirst.com/blog/brief-history-of-face-recognition-software/> [Accessed 25 Mar. 2018].

Reichert, C. (2017). Intel demos 5G facial-recognition payment technology | ZDNet. [online] ZDNet. Available at: <http://www.zdnet.com/article/intel-demos-5g-facial-recognition-payment-technology/> [Accessed 25 Mar. 2018].

Zhao, W., Chellappa, R., Phillips, P. and Rosenfeld, A. (2003). Face recognition. *ACM Computing Surveys*, 35(4), pp.399-458.

Pooja G.R, et al. (2010). An automated Attendance System Using Image Processing. *International Journal Of Advanced Networking & Applications*.

Wagh, P., Thakare, R., Chaudhari, J. and Patil, S. (2015). Attendance system based on face recognition using eigen face and PCA algorithms. *International Conference on Green Computing and Internet of Things*.

Arun Katara, Mr. Sudesh V. Kolhe, Mr. Amar P. Zilpe, Mr. Nikhil D. Bhele, Mr. Chetan J. Bele. (2017). "Attendance System Using Face Recognition and Class Monitoring System", *International Journal on Recent and Innovation Trends in Computing and Communication*, V5 (2).

S.Aanjanadevi, Dr.V.Palanisamy,R.Anandha Jothi. (2017). A Study on Secure Online Voting System using Biometrics Face Detection and Recognition Algorithms. *International Journal for Modern Trends in Science and Technology*, V3(8).

Wei-Lun Chao. (2007). Face Recognition, GICE, National Taiwan University. Akshara Jadhav. (2017). Automated Attendance System Using Face Recognition. *International Research Journal of Engineering and Technology*. V4 (1).

P. Arun Mozhi Devan et al., (2017). Smart Attendance System Using Face Recognition. *Advances in Natural and Applied Sciences*. 11(7), Pages: 139-144

Rahul V. Patil and S. B.Bangar. (2017). Video Surveillance Based Attendance system. *IJARCCCE*, 6(3), pp.708-713.

Mrunmayee Shirodkar. (2015). Automated Attendance Management System using Face Recognition. *International Journal of Computer Applications and International Conference and Workshop on Emerging Trends in Technology*.

